Assignment 2 320: Digital Therapy Assistant Solution Summary

Author

Name: Timmy Ngo (Core Services)

Group Members: Josh (Core Security), Trevor (AI/RAG)

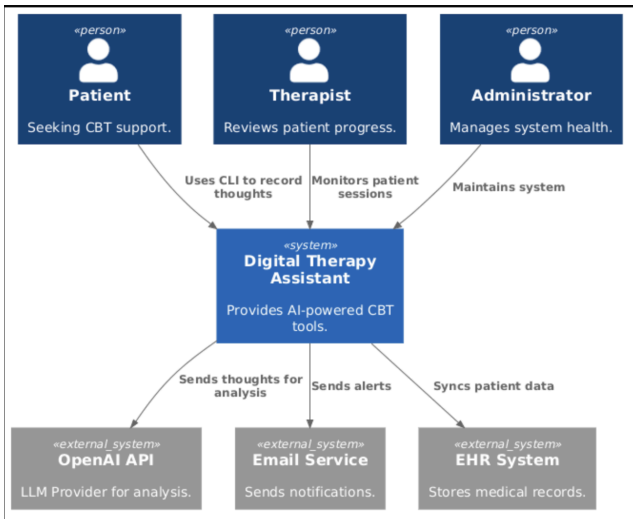
Email: timngo@chapman.edu, tlussier@chapman.edu, jomccarthy@chapman.edu

Date: 4/6/26

Diagrams

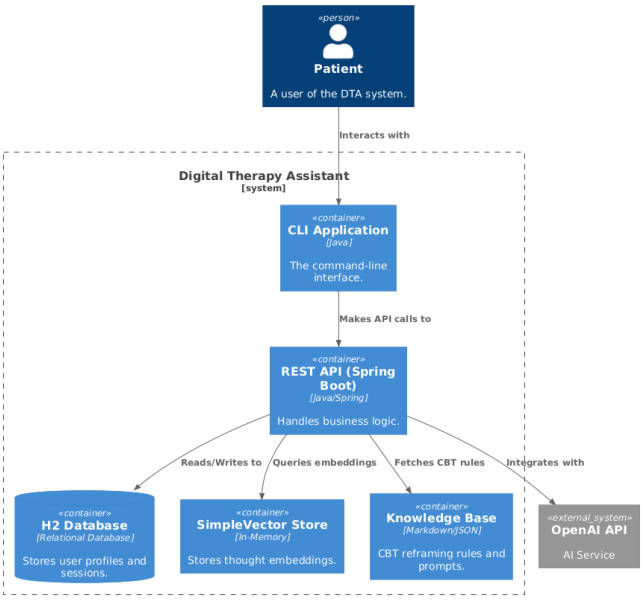
Context Diagram

Show Digital Therapy Assistant in context with users Patients, Therapists, Administrators) and external systems LLM Provider, Email Service, EHR Systems)



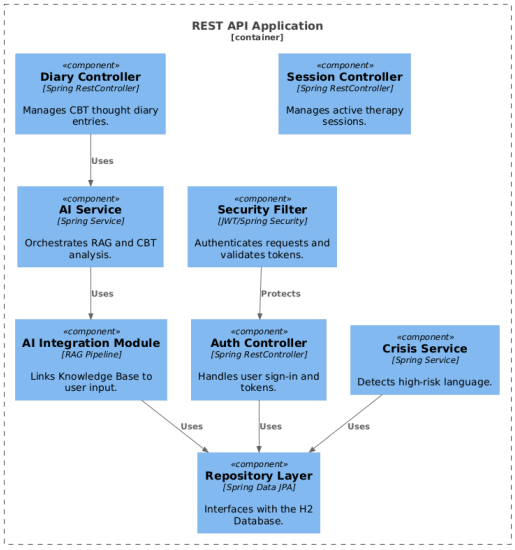
Container Diagram

Show high-level technical building blocks: CLI Application, Spring Boot Application, H2 Database, SimpleVector Store, Knowledge Base



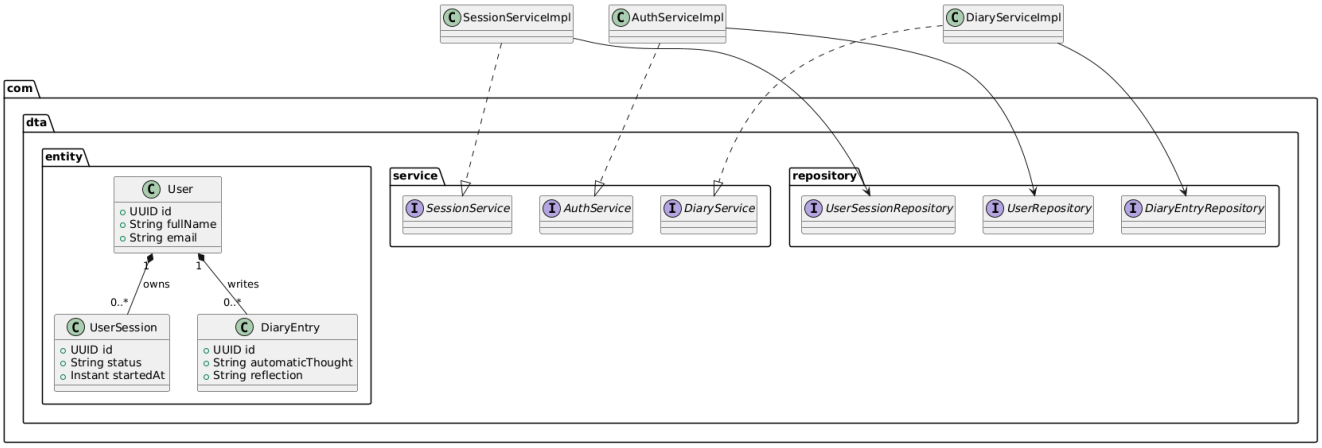
Component Diagram

Detail the Application components: Controllers, Services, Repositories, AI Integration modules, Security components



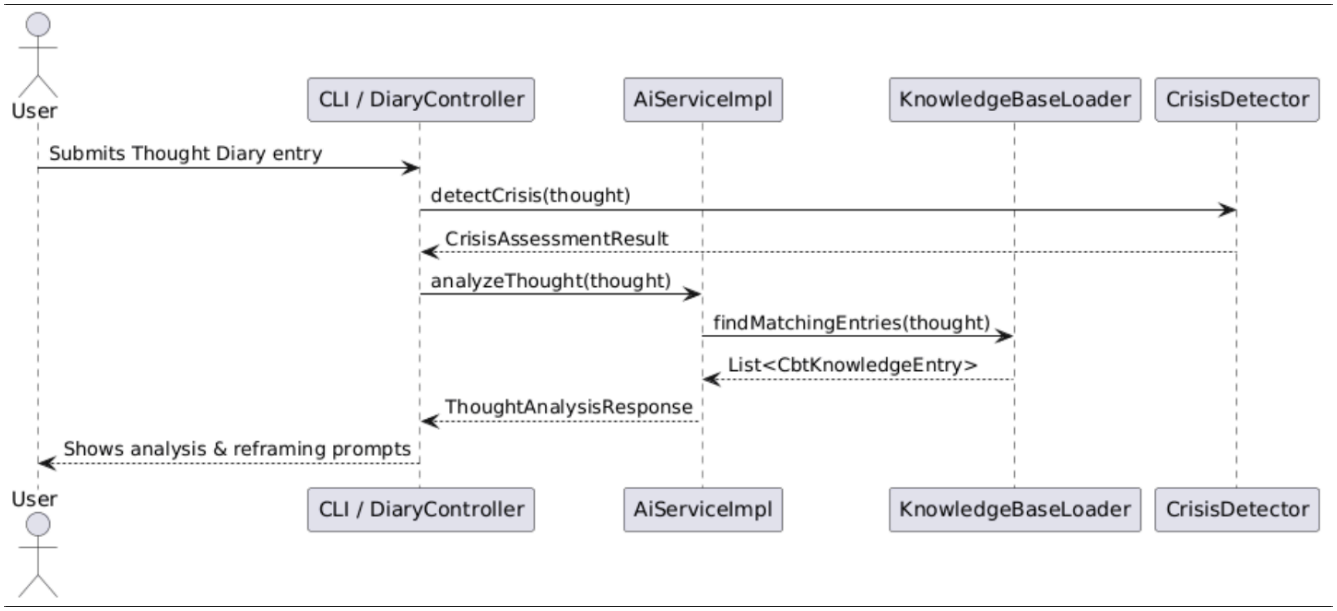
Class Diagram

Class Diagram describing System Classes



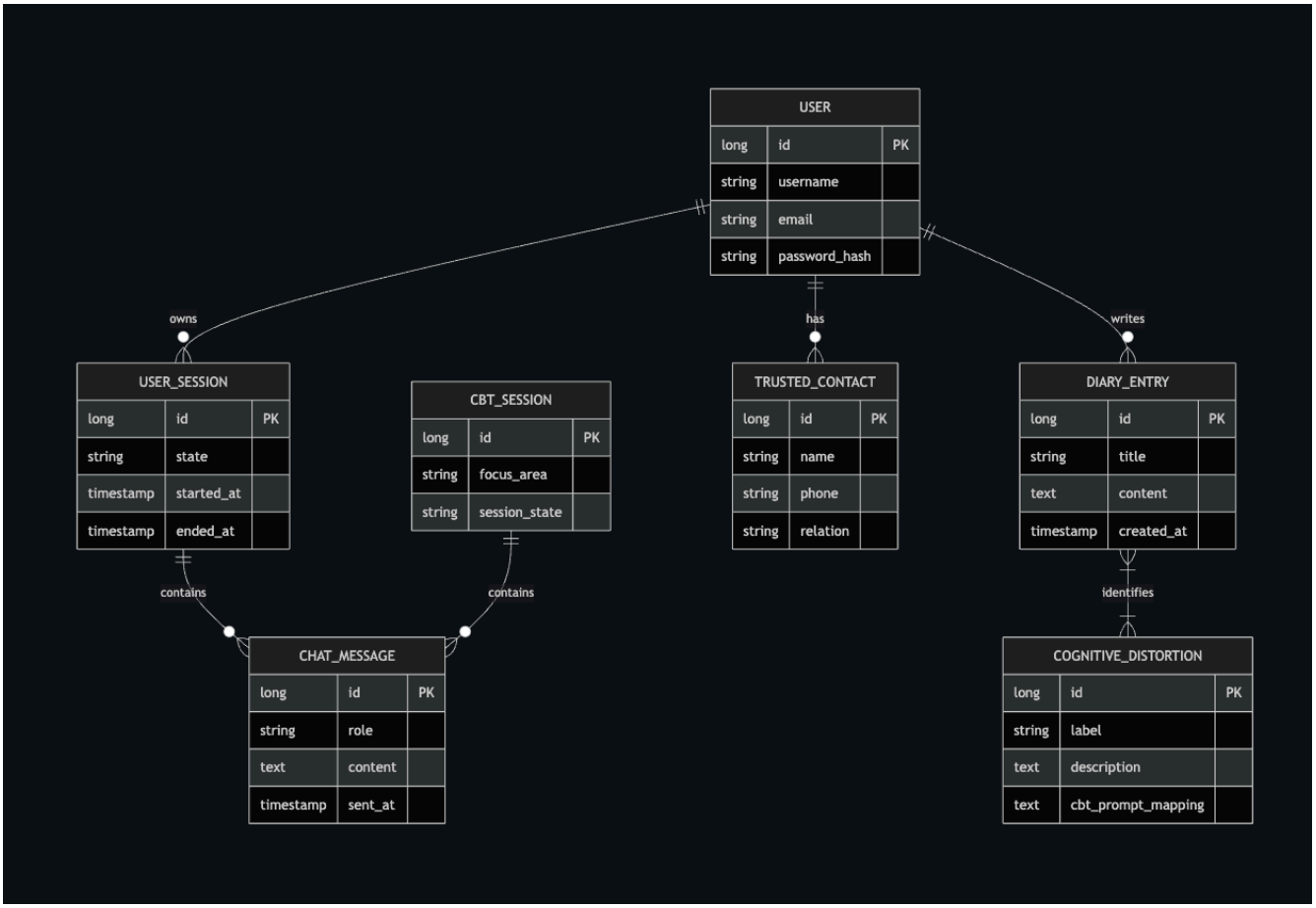
Sequence Diagram

Sequence Diagram describing Critical System Flow



ER Diagram

ER Diagram describing System Data Storage



Solution Contracts

This is a list of Service Contracts and how they are going to be met Ex. Will Implement OAuth2

- Security & Authentication: The system implements stateless authentication using JWT (JSON Web Tokens). This is managed through a JwtAuthFilter that validates tokens for every request.
- Data Persistence: The application uses Spring Data JPA to manage data storage. The contract is met using an H2 Relational Database with file-based storage to ensure data remains available between restarts.
- AI Service Integration: The system uses a Retrieval-Augmented Generation (RAG) pattern. It meets this contract by loading CBT knowledge from a local

KnowledgeBaseLoader and combining it with user input to generate context-aware reframing prompts.

- Crisis Safety: The system implements a Crisis Detection Contract that scans every message for high-risk keywords before processing. If a crisis is detected, it returns a specific immediate_intervention action and prevents standard AI processing.
- API Standards: All endpoints follow RESTful conventions, returning standard HTTP status codes (e.g., 201 Created for new entries) and using JSON for all request and response payloads.
- User Progress Tracking: The system calculates a dynamic Burnout Score based on activity metrics. This contract is met by aggregating the count of completed sessions and diary entries for a specific user ID.

API

Application Programming interface describing End Points for external System Access

#	End Point	API Method	Request Parameters	Response Parameters	Response Code	JIRA
1	/auth/register	POST	Full Name, Email, Password	JWT Access Token	201 Created	https://chapman-team-nux3f370.atlassian.net/browse/A13SA-4
2	/auth/login	POST	Email, Password	JWT Access Token	200 OK	https://chapman-team-nux3f370.atlassian.net/browse/A13S

						A-5
3	/diary/entries	POST	Thought, Emotion, Reflection	AI Analysis, Distortions	201 Created	https://chapman-team-nux3f370.atlassian.net/browse/A13SA-8
4	/progress/weekly	GET	User ID	Activity Score, Trends	200 OK	https://chapman-team-nux3f370.atlassian.net/browse/A13SA-11
5	/crisis/safety-plan	GET	User ID	Escalation Contacts	200 OK	https://chapman-team-nux3f370.atlassian.net/browse/A13SA-9

QA

Checklist, Test Cases, and a List of Unit, Integration Tests

#	Test Description	Assumptions	Test Data	Steps to Be Executed	Expected Result	JIRA
1	User Registration	Unique email	Valid user info	Send POST to /auth/register	User saved in H2; token returned	https://chapman-team-nux3f370.atlassian.net/browse/A13SA-13
2	Crisis Detection	AI service is up	"I want to hurt myself"	Submit thought via Diary service	isCrisis flag is set to TRUE	https://chapman-team-nux3f370.atlassian.net/browse/A13SA-16
3	CBT Reframing	KB is loaded	"I always fail at everything"	Pass thought to AI Analyzer	Returns "All-or-Nothing" distortion	https://chapman-team-nux3f370.atlassian.net/browse/A13SA-10
4	Secure Access	Valid JWT token	Bearer Token	Access /progress without login	Return 401 Unauthorized	https://chapman-team-nux3f370.atlassian.net/browse/A13S

						A-14
5	Progress Logic	Data exists in DB	User with 5 entries	Request weekly progress summary	Score calculated as "Improving"	https://chapman-team-nux3f370.atlassian.net/browse/A13SA-15

Jacoco Report Screenshot

4/8/26, 6:06 PM digital-therapy-assistant

digital-therapy-assistant Sessions

digital-therapy-assistant

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
com.dta.entity		37%		3%	83	132	104	181	69	118	4	9
com.dta.exception		7%		n/a	29	33	65	73	29	33	6	10
com.dta.security		0%		n/a	23	23	60	60	23	23	5	5
com.dta.controller		0%		n/a	24	24	36	36	24	24	5	5
com.dta.config		0%		n/a	13	13	26	26	13	13	2	2
com.dta.service.impl		87%		55%	22	65	22	228	8	47	0	6
com.dta.security.filter		0%		0%	8	8	24	24	2	2	1	1
com.dta.dto.response		76%		n/a	25	94	27	138	25	94	1	8
com.dta.ai		85%		44%	10	26	7	55	3	17	0	5
com.dta.dto.request		79%		n/a	9	44	13	62	9	44	1	8
com.dta.cli		71%		50%	5	11	6	30	2	7	0	2
com.dta		0%		n/a	2	2	3	3	2	2	1	1
com.dta.cli.command		81%		n/a	1	2	1	6	1	2	0	1
Total	1,374 of 3,082	55%	69 of 102	32%	254	477	394	922	210	426	26	63

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digital-therapy-assistant Sessions

digital-therapy-assistant

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com.dta.controller		0%		n/a	24	24	36	36	24	24	5	5
com.dta.config		0%		n/a	13	13	26	26	13	13	2	2
com.dta.service.impl		87%		55%	22	65	22	228	8	47	0	6
com.dta.security.filter		0%		0%	8	8	24	24	2	2	1	1
com.dta.dto.response		76%		n/a	25	94	27	138	25	94	1	8
com.dta.ai		85%		44%	10	26	7	55	3	17	0	5
com.dta.dto.request		79%		n/a	9	44	13	62	9	44	1	8
com.dta.cli		71%		50%	5	11	6	30	2	7	0	2
com.dta		0%		n/a	2	2	3	3	2	2	1	1
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